



11465 - Unveiling the dark matter structure of the least massive strong lensing galaxy ever found

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) J0040+2555

ABSTRACT

The nature of dark matter (DM) represents one of the most fundamental open questions in astrophysics. Although the cold dark matter (CDM) paradigm remains the prevailing framework, a number of alternative DM models have recently attracted increasing attention. Examining the spatial distribution of DM provides a promising means of constraining its nature, as different models predict distinct structural signatures. We serendipitously identified a remarkable system in which a $10^{8.5}$ solar mass dwarf galaxy, J0040-SL, acts as a strong lens. As the least massive strong lensing galaxy known, J0040-SL provides a unique laboratory to place the most stringent constraints on the DM profile at these scales. To fully exploit its uniqueness, we request coordinated JWST and HST imaging observations to precisely determine the stellar content of J0040-SL, a

critical input required to robustly disentangle DM from stars. These observations will enable a direct test of DM models in the dwarf galaxy regime.

OBSERVING DESCRIPTION

This program is to do NIRCам imaging to one local lensing system. NIRC observations are in five wide-band filters.

Proposal 11465 - Targets - Unveiling the dark matter structure of the least massive strong lensing galaxy ever found

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	J0040+2555	RA: 00 40 18.2600 (10.0760833d) Dec: +25 55 35.88 (25.92663d) Equinox: J2000		
	Comments: A bright star (TYC 1741-1325-1; V 11.4 mag, J 9.6 mag, H 9.1 mag, K 9.0 mag) is located approximately 15 arcseconds north of the target. Category=Galaxy Description=[Einstein rings, Elliptical galaxies] Extended=YES				

Proposal 11465 - Observation 1 - Unveiling the dark matter structure of the least massive strong lensing galaxy ever found

Observation	Proposal 11465, Observation 1									Tue Jun 09 21:00:38 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
	Comments: A bright star (TYC 1741-1325-1; V 11.4 mag, J 9.6 mag, H 9.1 mag, K 9.0 mag) is located approximately 15 arcseconds north of the target. To avoid contamination from its diffraction spikes and scattered light in the JWST observations, the telescope orientation should be chosen such that the diffraction spikes from this bright star do not intersect the science target. Forbidden V3PA values that would align spikes with north are 0°, 60°, 120°, 180°, 240°, 300°; please avoid these angles within ±10°.									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	J0040+2555	RA: 00 40 18.2600 (10.0760833d) Dec: +25 55 35.88 (25.92663d) Equinox: J2000							
	Comments: A bright star (TYC 1741-1325-1; V 11.4 mag, J 9.6 mag, H 9.1 mag, K 9.0 mag) is located approximately 15 arcseconds north of the target.									
	Category=Galaxy Description=[Einstein rings, Elliptical galaxies] Extended=YES									
Template	Module					Subarray				
	B					FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				5
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F070W	F277W	SHALLOW4	5	1	5	5	1288.412	274454
	2	F090W	F356W	SHALLOW4	5	1	5	5	1288.412	274454
	3	F115W	F410M	SHALLOW4	5	1	5	5	1288.412	274454
	4	F150W	F277W	SHALLOW4	5	1	5	5	1288.412	274454
	5	F200W	F444W	SHALLOW4	5	1	5	5	1288.412	274454
Special Requirements	Aperture PA Range 10.05262691 to 50.05262691 Degrees (V3 10.0 to 50.0) Aperture PA Range 70.05262691 to 110.05262691 Degrees (V3 70.0 to 110.0) Aperture PA Range 130.05262691 to 170.05262691 Degrees (V3 130.0 to 170.0) Aperture PA Range 190.05262691 to 230.05262691 Degrees (V3 190.0 to 230.0) Aperture PA Range 250.05262691 to 290.05262691 Degrees (V3 250.0 to 290.0) Aperture PA Range 310.05262691 to 350.05262691 Degrees (V3 310.0 to 350.0) Offset 38.0 arcsec, 38.0 arcsec									